

The Last Human Gate

Forward Deployed Engineering for Governance Automation

Jeremy Canale

contact@jeremycanale.com

<https://www.jeremycanale.com>

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Abstract

Enterprise governance requires decisions, evidence, and accountable authority; it does not require every review task to retain its current human implementation. We develop a task-substitution framework for Digital Governance Frameworks (DGF), treating each gate as an executable contract. Substitution requires sufficient accessible information, valid decision and authority checks, and a reduction in total human work after exceptions, verification, correction, and maintenance are counted. We derive a residual-work threshold and show why automating most cases can still increase labor. Forward deployed engineering connects these conditions to an architecture for agents, rule engines, evidence services, and escalation. DGF-Bench supplies controlled evidence from 300 synthetic projects and 899 evaluable model-project runs. Gemini 3.8 Flash, GPT-5.6 Luna, and DeepSeek v4.1 Flash achieve strict gate success of 94.98%, 83.29%, and 74.18%; complete-route success is 76.92%, 42.33%, and 24.67%. A deterministic control passes all 1,700 gates given the supplied rules and structured facts; this condition therefore establishes no incremental benefit from language-model inference. Evidence audits and 135 repeated runs distinguish correct decisions from reliable execution. A document counterexample establishes an information-sufficiency obstruction. These results support the technical feasibility of executing specified governance-review tasks in software; they do not estimate deployment labor savings. The framework makes workforce-reduction hypotheses testable through the human effort required at fixed output and quality. Sources, dossiers, traces, and analyses are public.

Keywords: task substitution; forward deployed engineering; AI agents; enterprise governance; human work; DGF-Bench.

1 Introduction: replacing execution, preserving the function

Before an enterprise buys software, connects systems, or releases an application, people review its architecture, security, contracts, cost, and operational readiness. The required outputs are decisions and their supporting work: identify a missing agreement, reject an unsafe network design, require a recovery test, or authorize a conditional release. These functions persist even if software performs work previously assigned to analysts, architects, and committee support staff. A professional title is an organizational allocation of tasks, not a technical specification of how those tasks must be executed.

This paper argues that a Digital Governance Framework (DGF), understood as an enterprise’s connected review gates, is a tractable candidate for early substitution of human review tasks by agentic systems. The claim concerns execution: an agent that produces a draft for someone to redo has supplied assistance; a system that completes an accepted review has performed that task. A workflow can contain substituted tasks and remaining human decisions at the same

time. The relevant system can combine language models, deterministic software, and escalation. Replacing every rule check with a language-model call is neither necessary nor desirable.

The DGF is attractive because it already organizes work into dossiers, policies, named authorities, bounded decisions, and recorded handoffs. This gives engineering a starting point that open-ended management lacks. The same features also reveal why a fluent agent is insufficient: facts may be missing, exceptions expensive, evidence unfaithful, and commitments unauthorized. Our claim is that these are explicit conditions of substitution, rather than reasons to assume that the current human task allocation must remain permanent. Whether DGF is earlier or easier to automate than every other business function is a comparative hypothesis, not an observed ranking.

We connect three contributions. First, a gate-contract formulation identifies information, validity, and authority requirements for delegating review execution. A source counterexample shows when no reviewer can satisfy a fixed decision contract from the permitted information. Second, a complete labor account gives a substitution threshold: reducing routine execution only reduces required human work if exception handling, verification, correction, and support remain below a computable budget. Third, DGF-Bench provides a public, controlled measurement of specified review tasks and their failure modes, including a deterministic control and repeated trajectories. The purpose of linking them is to make the route from an agent score to a workforce claim inspectable; none of the three substitutes for the others.

1.1 Position relative to prior work

Task-based accounts of automation distinguish displacement of existing work from demand changes and the creation of new tasks (Acemoglu and Restrepo, 2019). The persistence of difficult supervisory work after routine automation is a longstanding concern (Bainbridge, 1983). Sequential task assignment and quality complementarities further make handoffs and bottlenecks relevant (Demirer et al., 2026; Gans and Goldfarb, 2026). We apply these distinctions to the concrete boundary of a governance review, including cross-team handoffs and recurring engineering. The labor equations below are accounting identities and decision conditions, not a new macroeconomic theory or an employment forecast inferred from model accuracy.

Staged project gates predate agents (Cooper, 1990). Business-process compliance monitoring formalizes obligations and checks their execution (Ly et al., 2015); agentic process management extends the integration agenda (Calvanese et al., 2026). WorkArena measures enterprise interface tasks, ITBench operational IT scenarios, and τ -bench policy-guided tool interactions and repeated reliability (Drouin et al., 2024; Jha et al., 2025; Yao et al., 2024). DGF-Bench joins specialist review decisions into project routes and evaluates the supporting findings, actions, evidence, and authorization. Its percentages are not directly comparable with those benchmarks. Evidence-attribution work (Gao et al., 2023; Rashkin et al., 2023) motivates separating a correct answer from a supported one. Simple baselines and resource accounting matter when evaluating agents (Kapoor et al., 2024).

The contribution is this contract-to-workforce analysis supported by an inspectable review benchmark. It is narrower than a universal law of automation and more specific than a claim that language models improve productivity in general. This concise manuscript replaces the earlier empirical companion in the public repository and synthesizes the longer *The Last Human Gate* manuscript. Both versions use the same experimental record; they are not independent studies.

2 A gate as a substitutable contract

A DGF is an organization’s composition of review checkpoints. It is not a universal sequence: enterprises can merge, split, or reorder gates while preserving their obligations and information dependencies. Typical tasks include supplier due diligence, contractual checks, technical architecture review, security assessment, readiness verification, and final consolidation. We study three example routes, Buy, Integrate, and Build; the argument applies to the review contract, not to these particular names.

Definition 1 (Review contract). *A gate contract specifies permitted evidence and tools, a versioned policy, an authorization mandate, and the conditions for accepting an output. The output contains a disposition, findings, required actions, supporting evidence, and an authorization record. Its accepted form and substantive requirements are fixed before evaluating a proposed replacement.*

In DGF-Bench the dispositions are GO, GO_WITH_RESERVATIONS, REWORK, SUSPENSION, and NO_GO. A refusal can be a correct completed review. Conversely, an approval is unsuccessful if it omits a mandatory defect or exceeds the reviewer’s mandate. Completing the review is distinct from implementing the remediation it requests. The same boundary must apply to human and software work when estimating substitution.

2.1 Three conditions for a replacement

Information. The permitted observations must contain what the policy needs, or the contract must allow a request for information or abstention. A model cannot recover a missing fact merely by generating a plausible answer. Source accessibility matters as much as physical file existence; the agent may lack permission to read a relevant record.

Validity. The system must produce outputs satisfying the review’s quality and service requirements on its declared operating population. These include evidence support and the handling of uncertainty, not just a disposition matching a label. A machine-readable schema can reject malformed records but cannot by itself prove the substantive decision correct.

Authority. A valid mandate must permit the system’s action. An agent’s proposal and the mechanism that makes a commitment effective are separate. Deterministic enforcement can block forbidden commitments even when an agent requests them; a final authorized output does not establish that the model respected authority unaided. Delegating bounded execution does not eliminate the organization’s accountability or resolve deployment-specific legal requirements.

These conditions describe functional replacement. Net substitution of human work additionally requires a reduction in all human hours used to deliver that function, as derived next. Removing a routine reviewer while adding equivalent effort to project teams or support providers moves the work rather than reducing it.

2.2 An information obstruction

Let $I(x)$ denote all observations permitted in an evaluation condition for underlying case x , and $V(x)$ the set of outputs accepted by its fixed contract. Policy and mandate are held fixed.

Proposition 1 (Indistinguishable cases). *If two admissible cases satisfy $I(x_1) = I(x_2)$ but $V(x_1) \cap V(x_2) = \emptyset$, no system using only I can guarantee a valid output on both.*

Proof. Equal observations imply equal output distributions, including for a deterministic system. Guaranteed validity on both cases would require one distribution to assign probability one to both disjoint accepted sets, which is impossible. $\square$

This elementary observation is an operational test for an automation boundary, not a new general impossibility theorem. Our offline counterexample makes it concrete: two variants have identical extracted text in all 26 Word documents, but changing the underlying supplier due-diligence status, recorded in CSV evidence, changes the fixed reference decision from GO to REWORK. A condition restricted to those Word texts therefore cannot reproduce both reference decisions. Allowing the CSV, or changing the contract to accept a justified request for missing information, changes the task. For the fixed reference test, the accepted disposition sets are disjoint. This obstruction applies to a human reader as well as a model; it identifies a deficient information interface.

2.3 Why the DGF is a candidate

Table 1 turns the claim that DGF is suitable for automation into properties that can be checked in an organization. These are engineering advantages when present, not guarantees that every enterprise already has clean rules and accessible evidence.

Table 1: Properties that make a governance workflow a candidate for task substitution.

Property	Potential advantage	Condition that can defeat it
Recurring dossiers and outputs	Reusable extraction and decision interfaces	Material facts remain tacit or inaccessible
Explicit policies and mandates	Executable checks and bounded delegation	Conflicting rules or discretionary commitments
Recorded gate handoffs	Shared evidence and traceable dependencies	Work is repeated or lost between teams
Repeated review populations	Integration cost spread across cases	Low volume or frequent policy change
Observable acceptance criteria	Regression tests and monitored service	Evaluation rewards format instead of valid work

3 When agents replace human gate work

A complete account. At fixed case volume $\lambda > 0$, let $h_0 > 0$ be mean baseline handling hours. A frozen routing rule sends a fraction q of cases to substantive human exception handling. Let h_E be complete human hours on those cases, h_R ordinary-path human review, and h_W additional rework per case. Recurring human upkeep is B_H in the baseline and B_A after deployment. Then

$$L_H = \lambda h_0 + B_H, \quad L_A = \lambda \{q h_E + (1 - q) h_R + h_W\} + B_A. \quad (1)$$

Exceptions include their preliminary review; ordinary review includes mandatory signatures and expected sampled audits. Rework, downstream repair, supplier effort, evaluation, and policy maintenance are allocated once across the common process boundary. Changing a department or job title does not remove the hours. Initial implementation is reported separately for a transition-cost calculation; recurring adaptation belongs in B_A .

The fraction of cases retained by humans is generally different from the fraction of work retained. Let T be baseline handling time, J indicate exception routing, and $h_0 = \mathbb{E}[T]$. Define the residual’s baseline labor share $\phi = \mathbb{E}[TJ]/h_0$, its effort multiplier $\eta = h_E/\mathbb{E}[T \mid J = 1]$, and the ordinary group’s retained review fraction $\mu = h_R/\mathbb{E}[T \mid J = 0]$. Assume both conditional baseline means are positive and $0 < q < 1$; Equation (1) covers the other cases. With $r = h_W/h_0$ and $b_s = B_s/(\lambda h_0)$, the workload ratio is

$$\rho := \frac{L_A}{L_H} = \frac{\phi \eta + (1 - \phi) \mu + r + b_A}{1 + b_H}. \quad (2)$$

This is an accounting identity, not an estimated effect of agents.

Proposition 2 (Selection and the substitution frontier). *Let $T \geq 0$ be integrable with positive mean, and let $e(t) = \Pr(J = 1 \mid T = t)$. Under the definitions above,*

$$\phi = q + \frac{\text{Cov}(T, e(T))}{h_0}. \quad (3)$$

Thus routing that is nondecreasing in baseline effort retains at least as much baseline labor as its share of cases. For any target fractional workload reduction $d \in [0, 1]$, achieving $L_A \leq (1 - d)L_H$ is equivalent to

$$\phi\eta + (1 - \phi)\mu + r + b_A \leq (1 - d)(1 + b_H). \quad (4)$$

Proof. Conditional expectation gives $\mathbb{E}[TJ] = \mathbb{E}[Te(T)]$. Expanding the covariance and using $\mathbb{E}[e(T)] = q$ proves Equation (3). For independent copies T, T' , twice this covariance is $\mathbb{E}[(T - T')(e(T) - e(T'))]$, which is nonnegative when e is nondecreasing. Splitting h_0 between the two routing groups yields $qh_E/h_0 = \phi\eta$ and $(1 - q)h_R/h_0 = (1 - \phi)\mu$. Substitution into Equation (1) proves Equation (2); multiplication by its positive denominator gives the target condition. $\square$

The practical consequence is that an agent success rate cannot be read as the percentage of human labor removed. For example, 20% of cases retain 40% of baseline labor when their baseline handling time is twice the overall mean. If their post-deployment effort rises by 50%, those exceptions alone consume 60% of baseline handling hours. This is the mechanism by which a small exception queue can prevent a large workforce reduction.

Reliability consumes the same budget. If erroneous accepted outputs occur on a fraction p of all cases and cause additional human repair averaging h_C hours conditional on such an error, their contribution to r is ph_C/h_0 , counted once. This links execution reliability to the labor target, but does not make low repair cost a quality guarantee. An undetected harmful error is unacceptable even if it consumes no repair hours. Strict benchmark failures, human escalations, and erroneous accepted outputs are different events; the experiments do not measure deployment values of q or h_C .

Coverage does not determine required labor. The extended manuscript’s reproducible synthetic example allocates 140 baseline FTE across nine governance populations, at 120 useful hours per FTE per month. Each population has 100 pooled review visits, exception fractions from 0.16 to 0.33, and exception baseline means twice its overall mean. The workload-weighted exception fraction is 23.43%, while the groups retain 46.86% of baseline labor. With these partitions unchanged, the calculated requirements are shown in Figure 1. The scenario parameters (η, m, h_W, B_A) are respectively $(0.5, 0.10, 0, 0)$, $(1, 0.10, 0, 120)$, $(1, 0.10, 0.5, 120)$, and $(1.5, 0.35, 2, 240)$. Here $h_R = mh_0$, so m differs from the group-relative μ ; h_W is hours per case and B_A hours per population per month. Baseline upkeep is zero. The paired population vectors are baseline FTE $(10, 10, 30, 20, 10, 25, 15, 10, 10)$ and $q = (0.22, 0.33, 0.18, 0.16, 0.22, 0.24, 0.26, 0.33, 0.33)$. These are sensitivity calculations using declared assumptions, not measurements or forecasts. The comparison establishes a useful non-identification result: even fixed coverage and a fixed baseline case partition do not determine the sign of the labor effect. Their associated quality is not estimated. A deployment must measure both the contract outcomes and the complete account to identify whether execution has actually been substituted at acceptable quality. None of these four scenarios reaches an 80% reduction. With $\phi \simeq 0.4686$ and no other residual burden, that target already requires $\eta \leq 0.20/\phi \simeq 0.427$. Thus the forecast requires further reduction in exception effort or in its retained labor share, even before allowing for ongoing review and support.

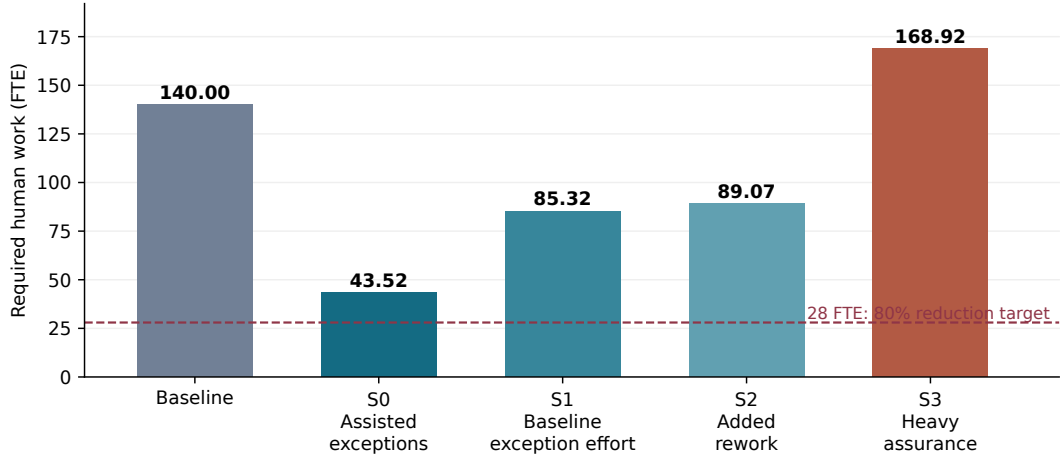


Figure 1: Illustrative human-work requirements with the same case partition: baseline 140 FTE; four operating assumptions produce 43.52, 85.32, 89.07, and 168.92 FTE. The dashed line is the author’s 80% reduction target, not a fitted forecast. Parameters and calculation are released.

From required work to staffing. At fixed useful capacity $K > 0$, required workload-equivalent FTE equal L/K . Under the additional assumptions of interchangeable workers, target utilization $0 < u \leq 1$, no separate coverage constraint, and staffing chosen to minimize labor cost, required headcount is $\lceil L/(uK) \rceil$. Reducing required hours then removes posts when this integer crosses a staffing threshold. Redeployment or increased demand can absorb the released capacity; they do not make execution of the old tasks indispensable again. For a finite collection of gates with bounded visit counts, vanishing human case effort leaves a support floor B_A/K ; zero total human labor further requires that this support work be eliminated or automated. These are explicit conditions for displacement, not a claim that current benchmark performance has already satisfied them.

4 The FDE implementation: engineering the whole workflow

Forward deployed engineers (FDEs) work alongside organizations to turn software capabilities into deployed systems; embedded implementation is explicit in the role described by Palantir Technologies (2026). Here the role has a concrete purpose: construct the information, policy, evidence, and authorization interfaces that let an accepted review be completed without requiring a person to reconstruct it. The job title itself does not establish an advantage.

Figure 2 separates responsibilities. Agents can investigate a dossier, extract candidate facts, interpret narrative material, and propose findings. Where policies are executable, a deterministic service evaluates them. Evidence services attach copied source values and locations rather than asking a model to reproduce quotations. Mandate checks govern commitments. The system escalates cases for which its evidence or delegated scope is inadequate. These are proposed deployment components, not a claim that this complete architecture was evaluated by the current benchmark.

The FDE’s work has four measurable deliverables. First, connect the required facts to accessible sources and demonstrate that the source set is sufficient for the declared task. Second, encode policy versions and mandates with the responsible owners, including the treatment of exceptions. Third, make the accepted record usable by the next gate without re-reading and re-verifying the whole dossier. Fourth, measure failures and human hours across the process, including the FDE’s own recurring work. Policy formalization and initial integration are transition investments; continued adaptation is an operating burden. Neither disappears because a supplier performs it.

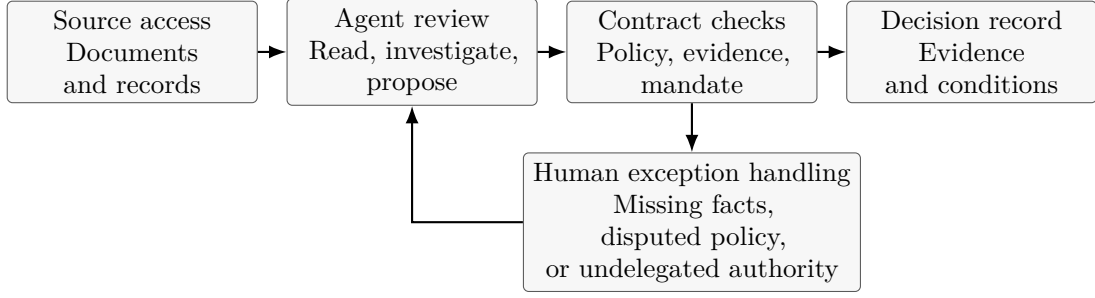


Figure 2: Proposed gate implementation. Every human contribution, including support of these services, belongs in the labor account. Passing a technical check alone is not a certificate of complete substantive correctness.

These deliverables explain the proposed DGF-first strategy. An FDE implementing an enterprise agent encounters project governance during deployment. The interfaces built to navigate it can also become interfaces through which its reviews execute. Policy ownership, evidence access, and repeated decision formats create reusable engineering opportunities. The claim is a plausible sequence of deployment work, not evidence that DGF must precede every business application. High-volume operational processes may justify automation earlier.

Designing common interfaces across gates matters even when rollout is incremental. Consider an Architecture review accepting a design only if a private endpoint is added. Security must receive both the condition and whether it has been fulfilled; receiving an unqualified approval would change the contract. A system that preserves such state can reduce repeated interpretation. A system that drops it can create new human verification work. Whole-route outcomes and the shared labor account test these alternatives; adding local automation percentages does not.

In deployment, the decisive object is therefore the *accepted task*, not the number of generated reports. Routine cases can have no required human execution while rare cases retain specialist review. If every case requires a lengthy signature or reconstruction, that review appears in h_R and can erase the gain. Conversely, keeping an accountable owner does not by itself imply keeping the same number of people executing each review.

5 Evidence: executing governance contracts

Design and scope. DGF-Bench tests the execution of a specified review task: inspect permitted evidence, choose a disposition, identify findings and required actions, cite the supporting observations, and respect the authorization contract. A seeded generator constructs 300 fictional projects, 100 each on Buy, Integrate, and Build routes. Buy follows Procurement, Legal, Compliance, Security, IT, and General; Integrate follows IT, Architecture, Security, Legal, Compliance, and General; Build follows IT, Architecture, Security, Tech Readiness, and General. This gives 1,700 gate occurrences per model. The dossiers include project charters and review requests, architecture diagrams, technical designs, supplier and legal records, and operational test evidence. Decision-coverage sampling balances applicable outcomes. Difficulty setting 4 controls generator parameters; it is not a validated scale of reasoning difficulty. The 300 projects have distinct architecture and complete decision-fact signatures, although some gate-level patterns recur.

The tested information condition supplies both executable policies and an authoritative structured **REVIEW_FACTS** snapshot. Models can read allowed evidence and use simulated enterprise tools; evaluator-only reference files are inaccessible. The model is the reviewer, not the generator of the project. Every gate returns one of **GO**, **GO_WITH_RESERVATIONS**, **REWORK**, **SUSPENSION**, or **NO_GO**. Correctly refusing a deficient proposal counts as success. Later gates receive the model’s actual

upstream reviews, and General consolidates the route.

Three exact OpenRouter endpoint identifiers were evaluated on 22–23 September 2026: `google/gemini-3.8-flash`, `openai/gpt-5.6-luna`, and `deepseek/deepseek-v4.1-flash`. Temperature was zero, vision was `auto`, and each gate permitted at most 20 turns, 40 tool calls, and 8,192 output tokens per turn. Of 900 planned model–project runs, 899 and their 5,094 gates are evaluable. One Gemini Integrate run failed at its first gate because of a provider generation error; its six gates are excluded. Recorded failed attempts remain in the cost accounting and released traces.

Outcomes and comparison. The frozen deterministic evaluator requires a correct disposition, exact finding and action sets, observed evidence references and exact observed field/value excerpts, and the correct authorization indicator. A *strict gate success* passes every component; a *complete route* passes every gate. Disposition accuracy alone therefore does not measure a complete review. Validated conditional approvals can change the effective reference outcome. A false approval proceeds when that effective reference requires rework, suspension, or refusal; a critical miss omits a finding classified as critical by the policy.

A deterministic control reads the same public snapshots and executes the public policy functions, then writes its predictions before a separate process invokes the frozen scorer. It does not read hidden references during generation, request optional conditional approvals, or extract facts from documents. It can quote full snapshots without the models’ per-turn output-token limit. Its perfect score establishes that the formalized review kernel is software-executable; this condition establishes no incremental benefit from language-model inference.

Table 2: Task execution on the original population and repeated trajectories. Counts preserve the original strict endpoint. Sensitivity accepts specified structural evidence matches; it is post-hoc and is not a semantic adjudication. The deterministic control was not repeated.

Outcome	Gemini	Luna	DeepSeek	Rules
Evaluable projects	299	300	300	300
Evaluable gates	1,694	1,700	1,700	1,700
Correct dispositions	1,694	1,668	1,619	1,700
Strict gate successes	1,609	1,416	1,261	1,700
Strict gate success (%)	94.98	83.29	74.18	100.00
Complete routes	230	127	74	300
Complete routes (%)	76.92	42.33	24.67	100.00
Evidence-only failures	85	221	338	0
False approvals / critical misses	0 / 0	1 / 3	1 / 1	0 / 0
Sensitivity: successful gates	1,678	1,454	1,313	—
Sensitivity: complete routes	284	144	85	—
<i>Follow-up: 15 existing dossiers, three trajectories per model</i>				
Strict gates / 255	245	211	187	—
Complete routes / 45	35	19	11	—
Dossiers passing all three / 15	9	3	0	—

What succeeds, and what still fails. Gemini reaches the correct effective disposition on every evaluable gate and satisfies all non-evidence components; its 85 strict failures concern evidence conformity. Its strict gate rate is 94.98% (95% interval 93.86–96.04), but only 76.92% of complete routes pass (71.82–81.34). Luna and DeepSeek also exhibit a large gate–route gap, alongside substantive errors: Luna approves one Legal review despite a missing data-processing agreement, and DeepSeek submits a placeholder `G0` at forced finalization when address overlap requires `NO_G0`. The model comparison therefore distinguishes successful disposition, complete

documented review, and dependable execution of a whole route.

Gate intervals resample whole cases within routes (2,000 draws, seed 81931); route intervals use Wilson bounds. They describe the generated population, retaining within-project dependence. They do not treat 5,094 gates as independent organizations. The common 299-case comparison preserves the ranking; its paired intervals and complete component tables are released.

An audit of all 690 gates with a failed evidence component leaves the primary scores unchanged and applies one declared sensitivity rule: accept exact relevant field/value subsets from one observed object despite field order or equivalent numeric serialization. No quote is repaired, no tool observation invented, and no join across different objects promoted. This recovers 69 Gemini, 38 Luna, and 52 DeepSeek gates (Table 2). Gemini’s rate becomes 99.06% at gate level and 94.98% at route level. The interpretation is specific: much of this model’s measured route failure depends on the evidence representation contract. The audit does not establish full premise coverage or independently validated semantic support.

A concrete delegated review. In DGF-BLD-035200_build, a generated vendor-portal project, Gemini’s Tech Readiness review finds a draft runbook. It names TR-OPS-001, proposes COMPLETE_OPERATIONAL_HANDOVER, and cites the observed value "runbook_status": "draft". After consulting an active mandate, it invokes the conditional-approval tool with a requirement to approve the runbook before production cutover. The tool executes the authorization but explicitly leaves the finding unresolved. The model submits GO_WITH_RESERVATIONS with the finding and action retained, satisfying the strict contract. This post-hoc example demonstrates a performed review and authorized workflow transition, not completed remediation. Authority resides in the surrounding system: across the original experiment, Gemini makes 466 rejected conditional-approval calls as well as 398 executed approvals. Its absence of scored false final approvals cannot be attributed solely to consistently correct initial action selection.

Repeatability and the remaining boundary. The follow-up fixes 15 existing dossiers, five per route, before collecting three new trajectories for every model-dossier pair: 135 completed runs and 765 gates. No best-of-three selection is used. Gemini, Luna, and DeepSeek pass respectively 35, 19, and 11 of 45 routes; only nine, three, and zero dossiers pass on all three trajectories. All follow-up false-approval and critical-miss counts are zero. Provider routing is not fixed, so these observations measure repeatability of the recorded system rather than intrinsic model randomness. Original and follow-up recorded API costs are USD 87.015829172 and USD 12.5598870228, totaling USD 99.58; development, local computation, human review, and deployment are not costed.

The information counterexample in Section 2 identifies a different failure boundary: some proposed document-only inputs cannot determine the required decision. Together, the observed executions and the source check show both an automatable formalized task and a specific obstruction that must be resolved before extending its scope.

6 From task substitution to a workforce hypothesis

The observations support a positive, bounded conclusion: software can execute the specified governance-review operations in this engineered environment. The deterministic control makes the case for an executable decision kernel especially clear. The agents demonstrate another implementation of many of these operations, with measurable differences in evidence handling, reliability, and cost. Neither result is weakened by the fact that software uses several kinds of components. What remains unresolved is the effort needed to construct and operate the interfaces when facts and policies arrive in less prepared forms.

This is also where the claim about human replacement becomes testable. If a deployed system performs accepted tasks previously executed by people and satisfies the labor inequality at unchanged output and quality, it has substituted for part of their work. Calling the remaining people supervisors does not restore the eliminated execution hours. At fixed demand, fewer required hours can support fewer posts; actual headcount also depends on allocation, utilization, redeployment, and new work. The argument therefore permits substantial workforce displacement without treating it as a measured consequence of this synthetic experiment.

6.1 The 2033 hypothesis and a falsifiable deployment test

The author’s original hypothesis is that the DGF ecosystem will require at least 80% fewer workload-equivalent FTE by 20 September 2033 than during the reference year ending on 20 September 2026, at comparable governed output, quality, and service. The boundary includes preparation, exceptions, verification, remediation of review errors, maintenance, supervision, and supplier support. On an illustrative 140-FTE baseline, the target is at most 28 FTE. This is an author-selected forecast, not an extrapolation fitted to benchmark scores.

The baseline window has already closed and no enterprise cohort has been enrolled. Testing this exact dated hypothesis requires auditable historical activity records and a declared population and aggregation rule before observing its endpoint. Otherwise it cannot be scored as confirmed. A newly registered baseline can support a separate seven-year prospective test, but cannot move the date of the original prediction. A credible test fixes the accounting boundary, useful-hours conversion, output mix, quality tolerances, and service levels. It reports the complete L_A/L_H account rather than headcount in a renamed department. A ratio above 0.20 at the deadline fails the proposed 80% target; achieving that ratio while degrading the fixed quality or service requirements also fails the substitution claim. Missing records make the result unassessable, not successful.

Full elimination of human DGF execution is a stronger, undated conjecture. It would require both case-level human effort and the recurring support floor to disappear. No finite benchmark here establishes that endpoint. The scientifically actionable question is how much accepted task execution can already be transferred and what prevents further transfer.

6.2 The next discriminating experiment

The most useful extension varies information preparation while preserving answerability. Compare agents receiving structured snapshots with agents receiving the source documents and operational records from which the same facts can be recovered. Separately vary executable versus natural-language policy. Fix source access, mandate, time and tool budgets, and the evidence contract before testing on held-out cases. Include the rule engine and any required extraction pipeline in the comparison. Report decision quality, supported findings, complete routes, abstention, and all engineering and operating costs.

This extension does not require claiming human-level accuracy. It asks which parts of a review the system can perform when the required facts are no longer supplied as a ready-to-use snapshot. Our counterexample means that simply deleting snapshots is not a valid design. The released source audit also identifies runtime tools that can return snapshots, so changing file visibility alone would not isolate the intended condition. The existing extraction ablation and General-only intervention are preparations, not additional measured results in this paper.

6.3 Limits of the present evidence

The benchmark’s policy, generator, and reference share an author-designed rule system; their agreement is internal consistency rather than validation of a universal business standard. Decision-

balanced synthetic cases do not estimate enterprise prevalence. There are three endpoints, one original trajectory per model and case, and repetitions on only 15 existing dossiers. The post-hoc structural evidence metric is not semantic adjudication. No measured human comparator, enterprise deployment, staffing effect, or comparative ranking of candidate business functions is available. These limits locate the evidence: execution of declared review contracts is measured; net labor substitution and the 2033 forecast remain hypotheses to test.

7 Conclusion

The last human gate is not defined by the last person to hold an approval title. It is the last task whose execution still requires human work after the evidence, policy, authority, and supporting services have been specified. A governance function can remain mandatory while software takes over tasks that previously required specialists to perform each review.

The DGF is a credible candidate for this transition because it already packages many of those tasks into repeated, documented decision interfaces. Our contribution makes the substitution claim conditional and measurable: sufficient accessible information, valid authorized outputs, and a complete human-work account. The model explains why broad case coverage can coexist with either a large reduction or an increase in required labor. The FDE implementation connects these conditions to evidence services, policy checks, mandates, handoffs, and exception handling.

DGF-Bench demonstrates substantial execution capability for specified review tasks and exposes the limits of inferring workflow reliability from correct decisions. Its deterministic control also shows that a formalized review kernel can be executed without a language model. The consequent research question is where agents can extend that kernel into less formalized work at acceptable quality and lower total human effort. If they do so, replacement of human task execution is the result, not merely an improvement in report drafting. Whether that reaches the author’s 80% DGF-FTE reduction by 2033 requires the separate observations and fixed boundary specified here.

Data, code, and attribution

Jeremy Canale is the author: contact@jeremycanale.com, <https://www.jeremycanale.com>, <https://www.linkedin.com/in/jcanale13>. The public repository is <https://github.com/jeremy1392/dgf-agentic-bench>. Its `dgf-bench-300-20260923` release contains the source snapshot, complete generated dossiers including Word documents and architectures, model traces, failed attempts, scores, and cost ledgers. Original results and offline reproduction are in `research/2026-09-dgf-bench/`; controls, all-model evidence audits, repetitions, and source counterexamples are in `research/2026-09-followup/`. Use the frozen experimental source to reproduce the collected run. Synthetic workforce parameters are in `paper/anc/parameters.json`; this concise manuscript and its verification script are in `paper2/`. The extended manuscript remains in `paper/`. Earlier empirical-companion versions remain in repository history. No new model calls or human studies were conducted for this synthesis. The author is responsible for the claims, artifacts, and interpretation.

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